| Activity No. <3.1> | |
|---|---|
| <Control Structures Part 2> | |
| **Course Code:** CPE 007 | **Program:** Computer Engineering |
| **Course Title:** PROGRAMMING LOGIC AND DESIGN | **Date Performed:16/08/2025** |
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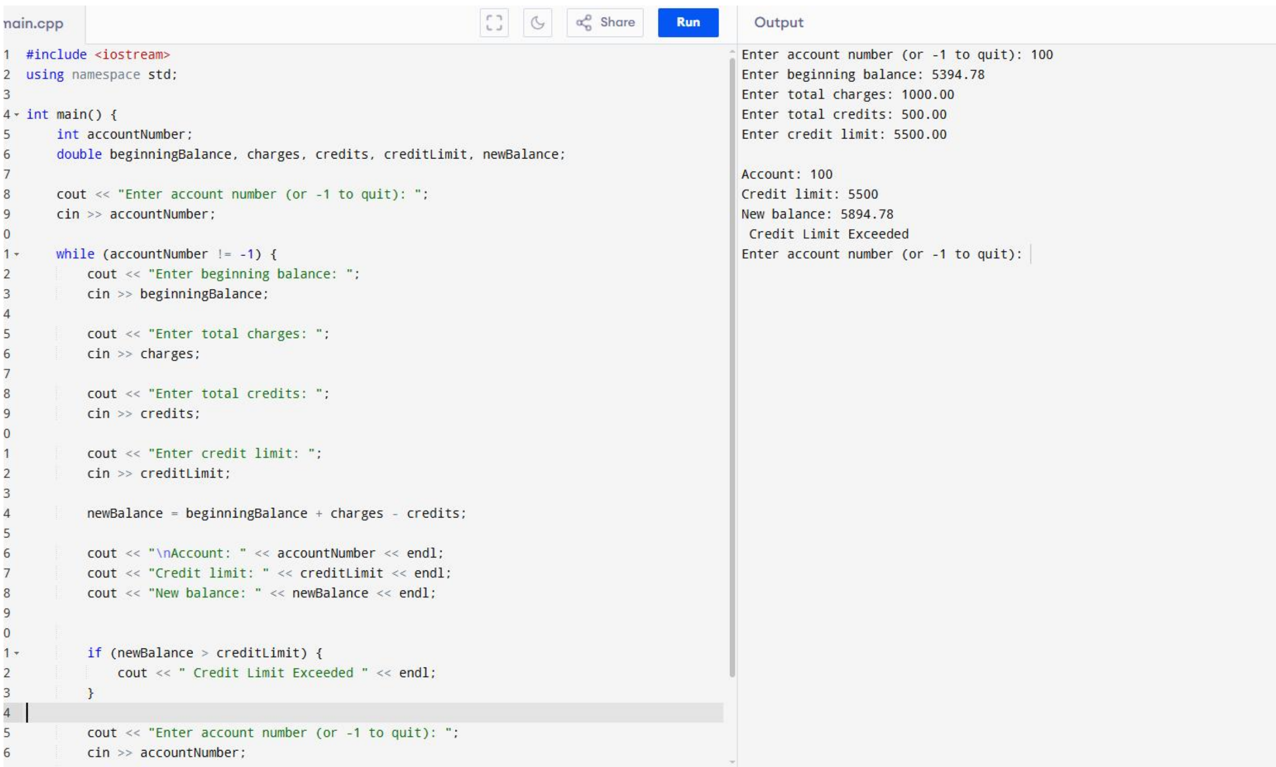
## 6. Output

1. **Develop a C++ program that will determine if a department store customer has exceeded the credit limit on a charge account. For each customer, the following facts are available:**




```cpp
#include <iostream>
using namespace std;

int main() {
    int accountNumber;
    double beginningBalance, charges, credits, creditLimit, newBalance;

    cout << "Enter account number (or -1 to quit): ";
    cin >> accountNumber;

    while (accountNumber != -1) {
        cout << "Enter beginning balance: ";
        cin >> beginningBalance;

        cout << "Enter total charges: ";
        cin >> charges;

        cout << "Enter total credits: ";
        cin >> credits;

        cout << "Enter credit limit: ";
        cin >> creditLimit;

        newBalance = beginningBalance + charges - credits;

        cout << "\nAccount: " << accountNumber << endl;
        cout << "Credit limit: " << creditLimit << endl;
        cout << "New balance: " << newBalance << endl;


        if (newBalance > creditLimit) {
            cout << " Credit Limit Exceeded " << endl;
        }

        cout << "Enter account number (or -1 to quit): ";
        cin >> accountNumber;
```

```
Enter account number (or -1 to quit): 100
Enter beginning balance: 5394.78
Enter total charges: 1000.00
Enter total credits: 500.00
Enter credit limit: 5500.00

Account: 100
Credit limit: 5500
New balance: 5894.78
 Credit Limit Exceeded
Enter account number (or -1 to quit):
```

2. **Because of the price of gasoline, drivers are concerned with the mileage obtained by their automobiles. One driver has kept track of several tankfuels of gasoline by recording miles driven and gallons used for each tankful. Develop a program that will input the miles driven and gallons used for each tankful. The program should calculate and display the miles per gallon obtained for each tankful. After processing all input information, the program should calculate and print the combined miles per gallon obtained for all tank fuels .**

main.cpp

```cpp
#include <iostream>
 using namespace std;

int main () {

     double gallons, miles, totalMiles =0, totalGallons = 0;
     int count=0;

      while (true) {
         cout<<"Enter the gallons used (-1 to end):";
         cin>> gallons;
         if ( gallons == -1)
         break;

         cout<<"Enter the miles driven:";
         cin>> miles;

         totalMiles += miles;
         totalGallons += gallons;
         double miles_per_gallon = miles / gallons ;
         cout<<"the miles/gallon for this was:" << miles_per_gallon <<endl;
         count ++;
      }
      if (count > 0 ){
         double average = totalMiles/totalGallons ;

         cout<<"the overall average miles//gallon was" << average <<endl;


        return 0;
      }
```

Output

```
Enter the gallons used (-1 to end):12.8
Enter the miles driven:287
the miles/gallon for this was:22.4219
Enter the gallons used (-1 to end):10.3
Enter the miles driven:200
the miles/gallon for this was:19.4175
Enter the gallons used (-1 to end):5
Enter the miles driven:120
the miles/gallon for this was:24
Enter the gallons used (-1 to end):-1
the overall average miles//gallon was21.6014

=== Code Execution Successful ===
```

**3. Create a program that will calculate the cost of sending a small parcel. The post office charges P5.00 for the first 300g, and P2.00 for every 100g thereafter (rounded up), up to a maximum weight of 1000g.**

**Succesful:**

main.cpp

```cpp
#include <iostream>
#include <cmath>
   using namespace std;

int main() {
    double weight, cost;

    cout << "Enter parcel weight (g): ";
    cin >> weight;

    if (weight <= 0) {
        cout << "Invalid weight." << endl;
    }
    else if (weight > 1000) {
        cout << "Parcel exceeds maximum weight limit (1000g)." << endl;
    }
    else {
        if (weight <= 300) {
            cost = 5.00;
        } else {

            double extraWeight = weight - 300;

            int extraBlocks = ceil(extraWeight / 100.0);
            cost = 5.00 + (extraBlocks * 2.00);
        }

        cout << "Parcel weight: " << weight << "g" << endl;
        cout << "Total cost: P" << cost << endl;
    }

    return 0;
}
```

Output

```
Enter parcel weight (g): 800
Parcel weight: 800g
Total cost: P15

=== Code Execution Successful ===
```

## Limit

main.cpp

```cpp
#include <iostream>
#include <cmath>
   using namespace std;

int main() {
    double weight, cost;

    cout << "Enter parcel weight (g): ";
    cin >> weight;

    if (weight <= 0) {
        cout << "Invalid weight." << endl;
    }
    else if (weight > 1000) {
        cout << "Parcel exceeds maximum weight limit (1000g)." << endl;
    }
    else {
        if (weight <= 300) {
            cost = 5.00;
        } else {

            double extraWeight = weight - 300;

            int extraBlocks = ceil(extraWeight / 100.0);
            cost = 5.00 + (extraBlocks * 2.00);
        }

        cout << "Parcel weight: " << weight << "g" << endl;
        cout << "Total cost: P" << cost << endl;
    }

    return 0;
}
```

Output

```
Enter parcel weight (g): 1200
Parcel exceeds maximum weight limit (1000g).


=== Code Execution Successful ===
```

**4. Write a program that displays a menu for simple conversion such as the following:**

main.cpp

```cpp
#include <iostream>
  using namespace std;

    int main() {

          int cm;
             int choice;
              float value, result;
               char again;

           do {

                cout<<"1. cm - inches"<<endl;
                cout<<"2. inches - cm"<<endl;
                 cout<<"3.feet - meter"<<endl;
                 cout<<"4.meter feet"<<endl;
                 cout<<"Enter ( 1 - 4):"<<endl;
                 cin>> choice;

                 switch (choice) {
                 case 1:
                      cout<<"Enter value in cm:";
                      cin>> value;
                      result = value / 2.54;
                      cout<<value << "cm = " << result << "inches"  <<endl;
                      break;

                 case 2:
                      cout<<"Enter value in inches:";
                      cin>> value;
                      result = value * 2.54;
                      cout<< value << "inches = " <<result << "cm" <<endl;
                      break;
                 case 3:
                      cout<<"Enter value in meter:";
                      cin>> value;
```

Output

```
1. cm - inches
2. inches - cm
3.feet - meter
4.meter feet
Enter ( 1 - 4):
1
Enter value in cm:36
36cm = 14.1732inches

Do you want to convert another? (Y/N): y
1. cm - inches
2. inches - cm
3.feet - meter
4.meter feet
Enter ( 1 - 4):
2
Enter value in inches:94
94inches = 238.76cm

Do you want to convert another? (Y/N): y
1. cm - inches
2. inches - cm
3.feet - meter
4.meter feet
Enter ( 1 - 4):
3
Enter value in meter:54
54meter = 16.4592feet
Do you want to convert another? (Y/N): y
1. cm - inches
2. inches - cm
3.feet - meter
4.meter feet
Enter ( 1 - 4):
4
Enter value in feet:345
```

```
4.meter feet
Enter ( 1 - 4):
4
Enter value in feet:345
345feet = 1131.89meter
Do you want to convert another? (Y/N): n

Thank you!


=== Code Execution Successful ===
```

**5. Write a program that displays a menu for simple computation of formula such as the following:**

main.cpp

Share Run

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int choice;
    float radius, length, width, base, height, side, area;
    char again;

    do {

        cout << "1. Area of Circle" << endl;
        cout << "2. Area of Rectangle" << endl;
        cout << "3. Area of Triangle" << endl;
        cout << "4. Area of Square" << endl;
        cout << "Enter your choice (1-4): ";
        cin >> choice;

        switch(choice) {
            case 1:
                cout << "Enter radius: ";
                cin >> radius;
                area = M_PI * radius * radius;
                cout << "Area of Circle = " << area << endl;
                break;

            case 2:
                cout << "Enter length: ";
                cin >> length;
                cout << "Enter width: ";
                cin >> width;
                area = length * width;
                cout << "Area of Rectangle = " << area << endl;
                break;

            case 3:
```

Output

```
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 1
Enter radius: 23
Area of Circle = 1661.9

Do you want to compute another area? (Y/N): y
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 2
Enter length: 43
Enter width: 55
Area of Rectangle = 2365

Do you want to compute another area? (Y/N): y
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 3
Enter base: 45
Enter height: 66
Area of Triangle = 1485

Do you want to compute another area? (Y/N): y
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 4
Enter side: 43
Area of Square = 1849
```

main.cpp

Share Run

```cpp
                cin >> length;
                cout << "Enter width: ";
                cin >> width;
                area = length * width;
                cout << "Area of Rectangle = " << area << endl;
                break;

            case 3:
                cout << "Enter base: ";
                cin >> base;
                cout << "Enter height: ";
                cin >> height;
                area = 0.5 * base * height;
                cout << "Area of Triangle = " << area << endl;
                break;

            case 4:
                cout << "Enter side: ";
                cin >> side;
                area = side * side;
                cout << "Area of Square = " << area << endl;
                break;

            default:
                cout << "Invalid choice! Please select 1-4." << endl;
        }

        cout << "\nDo you want to compute another area? (Y/N): ";
        cin >> again;

    } while (again == 'Y' || again == 'y');

    cout << "\nThank you!" << endl;
    return 0;
}
```

Output

```

Do you want to compute another area? (Y/N): y
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 2
Enter length: 43
Enter width: 55
Area of Rectangle = 2365

Do you want to compute another area? (Y/N): y
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 3
Enter base: 45
Enter height: 66
Area of Triangle = 1485

Do you want to compute another area? (Y/N): y
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
Enter your choice (1-4): 4
Enter side: 43
Area of Square = 1849

Do you want to compute another area? (Y/N): n

Thank you!


=== Code Execution Successful ===
```

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|---|
| |
| **7. Supplementary Activity** |
| |
| **8. Conclusion** |
| In this coding I actually understand the  if,else, while, better and how to use them correctly, and the DOUBLES, when I first try to write this code, I didn't know how to use double propperly then I find out that its juts a circle that you can put them together to make the code easier to read, and for now im just a little bit confused in the switch and I need to learn how it works propperly, though I can use them but I feel like sometimes im a little bit lost but I know soon that theres more knowledge to come. |
| **9. Assessment Rubric** |
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